Nnamdi Azikiwe University, Awka.
Department of Mathematics.

First Semester Examination for 2021/2022 Session, **MAT 201**: Linear Algebra I

Instruction: Shade the correct option in the boxes using BIRO. Double shading and mutilations attracts zero mark. Mutilation of registration number attracts no result. NOTA is an acronym for None of the above.

Answer all questions **Time: 1Hr 30 Mins**

Name(s):_____ **RegNumber**_____

Department:_____ **Signature:**_____

1	A	B	C	D	E	11	A	B	C	D	E	21	A	B	C	D	E
2	A	B	C	D	E	12	A	B	C	D	E	22	A	B	C	D	E
3	A	B	C	D	E	13	A	B	C	D	E	23	A	B	C	D	E
4	A	B	C	D	E	14	A	B	C	D	E	24	A	B	C	D	E
5	A	B	C	D	E	15	A	B	C	D	E	25	A	B	C	D	E
6	A	B	C	D	E	16	A	B	C	D	E	26	A	B	C	D	E
7	A	B	C	D	E	17	A	B	C	D	E	27	A	B	C	D	E
8	A	B	C	D	E	18	A	B	C	D	E	28	A	B	C	D	E
9	A	B	C	D	E	19	A	B	C	D	E	29	A	B	C	D	E
10	A	B	C	D	E	20	A	B	C	D	E	30	A	B	C	D	E

1. Given that the sides of an equilateral triangle is represented by the vectors $\bar{u}$, $\bar{v}$, and $\bar{w}$. If the vector $\bar{u}$ is 2 units, find the $\bar{u} \cdot \bar{v}$
   A. 4 B. -4 C. 2 D. -2 E. NOTA
2. A linear equation in the variable x and y, whose general solution is $x = 5 + 2t$ and $y = t$ is
   A. $x + 2y = 5$ B. $x - 2y = 5$ C. $2x - y = 5$ D. $x + y = 5$ E. NOTA
3. At which point does the line with parametric equation $x = -1 - 7t, y = 2 + 3t, z = 3 - 2t$ intersect the yz-plane
   A. $\left(0, \frac{11}{7}, \frac{-23}{7}\right)$ B. $\left(0, \frac{-11}{7}, \frac{23}{7}\right)$ C. $\left(0, \frac{7}{11}, \frac{7}{23}\right)$ D. $\left(0, \frac{11}{7}, \frac{23}{7}\right)$
   E. NOTA
4. Find a unit vector normal to the plane $3x - 4y + 12z + 7 = 0$
   A. $\frac{1}{13}(3, -4, 12)$ B. $\frac{1}{\sqrt{13}}(3, -4, 12)$ C. $\frac{1}{13}(3, 4, 12)$ D. $13(3, 4, 12)$ E. NOTA
5. Given that the standard matrix for the linear transformation T is $\begin{bmatrix} 1 & 2 \\ 4 & 0 \end{bmatrix}$, then find $T(-1,2)$
   A. (3,4) B. (4,3) C. $(-3,4)$ D. $(-3, -4)$ E. $(3, -4)$
6. At what point will the line whose equation $(2 + 3t)i - 4tj + (5 + t)k$ intersect the plane $4x + 5y - 2z = 18$.
   A. $(-4, 8, 3)$ B. $(4, 8, 3)$ C. $(4, 8, -3)$ D. $(4, -8, 3)$ E. $(-4, -8, -3)$
7. The position vectors of P and Q are $(1, -2, 4)$ and $(4, -5, -2)$ respectively. Find $\overrightarrow{PQ}$
   A. $(3, 3, 6)$ B. $(-3, -3, -6)$ C. $(3, -3, -6)$ D. $(-3, 3, 6)$ E. NOTA
8. Find the vector projection of $\bar{v}$ onto $\bar{u}$, where $\bar{u} = -i - 2j + 2k$ and $\bar{v} = 3i + 3j + 4k$
   A. $\frac{2}{9}i + \frac{2}{9}j + \frac{1}{9}k$ B. $\frac{1}{9}i + \frac{2}{9}j - \frac{2}{9}k$ C. $\frac{1}{9}i - \frac{2}{9}j + \frac{2}{9}k$ D. $-\frac{1}{9}i - \frac{2}{9}j + \frac{2}{9}k$ E. NOTA

66
13
7

Assignment

9. Find the direction cosine of the vector $\bar{u} = i + 2j - 2k$

A. $\left(\frac{1}{3}, \frac{2}{3}, \frac{2}{3}\right)$ B. $\left(\frac{1}{9}, \frac{2}{9}, -\frac{2}{9}\right)$ C. $\left(\frac{1}{3}, \frac{2}{3}, -\frac{2}{3}\right)$ D. $\left(-\frac{1}{3}, \frac{2}{3}, \frac{2}{3}\right)$ E. NOTA

10. Find the dot product of the two vectors given by $\bar{p} = 3i - 4j + 5k$ and $\bar{q} = i - 2j + 3k$

A. 12 B. 26 C. -26 D. 34 E. NOTA

11. Given that the matrix A is a 5×5 matrix whose determinant is -12, then $\det(2A) =$ .....

A. -24 B. 24 C. -384 D. -48 E. -364

12. The value of k so that $\bar{u}$ and $\bar{v}$ are orthogonal, where $\bar{u} = (k, -2, 3)$ and $\bar{v} = (3, 6, 3)$ is

A. 3 B. -1 C. 1 D. 2 E. None

13. A unit vector in the direction of v, where $v = (3,4)$ is

A. $\left(\frac{3}{4}, \frac{4}{3}\right)$ B. (3,4) C. $\left(\frac{3}{5}, \frac{4}{5}\right)$ D. $\left(\frac{3}{25}, \frac{4}{25}\right)$ E. NOTA

14. If $A = \begin{pmatrix} 2 & 2 \\ x & 4 \end{pmatrix}$, find the value of x for which the matrix is singular

A. 4 B. 2 C. 1 D. 0 E. NOTA

15. Which vector is perpendicular to the two vectors $\bar{u} = (1, 0, -2)$ and $\bar{v} = (3, -3, 1)$

A. $6i - 7j - 3k$ B. $-6i - 7j - 3k$ C. $6i - 7j + 3k$ D. $i + 5j - 2k$ E. NOTA

16. Find the modulus of $i \sin\theta + j \cos\theta$

A. 2 B. $\sin\theta$ C. $\cos\theta$ D. 1 E. NOTA

17. If $P = (1, -5, 2)$ and $Q = (-4, 0, -3)$ find the magnitude of $\overline{PQ}$

A. 25 units B. $\sqrt{50}$ units C. $3\sqrt{5}$ units D. $5\sqrt{3}$ units E. NOTA

18. Given that $A = \begin{pmatrix} 3 & 0 \\ 1 & 1 \end{pmatrix}$, find A^T

A. $\begin{pmatrix} 3 & 0 \\ 1 & 1 \end{pmatrix}$ B. $\begin{pmatrix} 3 & 1 \\ 0 & 1 \end{pmatrix}$ C. $\begin{pmatrix} 1 & 1 \\ 3 & 0 \end{pmatrix}$ D. $\begin{pmatrix} 0 & 1 \\ 1 & 3 \end{pmatrix}$ E. NOTA

19. Find the inverse of the matrix $A = \begin{pmatrix} 7 & 3 \\ 4 & -1 \end{pmatrix}$

A. $\frac{1}{19}\begin{pmatrix} 7 & 3 \\ 4 & -1 \end{pmatrix}$ B. $\frac{1}{19}\begin{pmatrix} 1 & 4 \\ 3 & -7 \end{pmatrix}$ C. $\frac{-1}{8}\begin{pmatrix} 1 & 3 \\ 4 & 5 \end{pmatrix}$ D. $\frac{-1}{19}\begin{pmatrix} -1 & -3 \\ -4 & 7 \end{pmatrix}$ E. NOTA

20. Find the scalar projection of the vector $\bar{a} = i - 2j + k$ on the vector $\bar{b} = 4i - 4j + 7k$

A. $\frac{9}{19}$ B. $\frac{21}{9}$ C. $\frac{-19}{9}$ D. $\frac{23}{9}$ E. $\frac{19}{9}$

21. Which of the following is not a linear combination of $A = \begin{pmatrix} 1 & 4 \\ 3 & 0 \end{pmatrix}$ and $B = \begin{pmatrix} 2 & 0 \\ 1 & 2 \end{pmatrix}$

A. $\begin{pmatrix} 1 & 12 \\ 8 & -2 \end{pmatrix}$ B. $\begin{pmatrix} 3 & 4 \\ 4 & 2 \end{pmatrix}$ C. $\begin{pmatrix} 3 & 4 \\ 1 & 4 \end{pmatrix}$ D. $\begin{pmatrix} 3 & -4 \\ -1 & 4 \end{pmatrix}$ E. NOTA

22. Equation of the plane that passes through the point $P(1,2,-1)$, with normal vector $\bar{v} = (9,1,-5)$ is

A. $x - 5y = 6$ B. $9x + 2y - 5z = 0$ C. $9x + y - 5z = 16$ D. $9x - y + 5z = 16$ E. NOTA

23. Find the angle between the vectors $\bar{a} = 4i + 2j - 2k$ and $v = i + j + 3k$

A. 60 B. 90 C. 0 D. 45 E. 180

24. The values of the minors of the elements of the first row of the matrix $\begin{pmatrix} 2 & -1 & 4 \\ 4 & 2 & -3 \\ 1 & 1 & 2 \end{pmatrix}$ are

A. 7, 11, 2 B. 2, 7, 11 C. 11, 2, 7 D. 7, -11, 2 E. NOTA

25. The values of the cofactor of the elements -3 in the second row of the matrix $\begin{pmatrix} 2 & -1 & 4 \\ 4 & 2 & -3 \\ 1 & 1 & 2 \end{pmatrix}$ is

A. 3 B. 2 C. 4 D. -3 E. -1